



Student No.

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Student Name: _____

Teacher Name: _____

2019

YEAR 12 / HSC - 2020
ASSESSMENT TASK 1

Mathematics Advanced

General Instructions

- Reading time – 10 minutes
- Working time – 80 minutes
- Write using black pen
- Approved calculators may be used
- A reference sheet is provided
- For Questions in Section II, show relevant mathematical reasoning and/or calculations

Total marks:
45

Section I – 5 marks – Pages 3 - 4

- Attempt Questions 1-5
- Allow about 10 minutes for this section

Section II – 40 marks – Pages 6 - 14

- Attempt Questions 6 - 14
- Allow about 1 hours and 10 minutes for this section

SECTION 1

5 MARKS

Attempt questions 1 – 5

Allow about 10 minutes for this section

Use the multiple-choice answer sheet for questions 1-5

1. If $f(x) = \log_e(2x)$ then find the value of $f'(1)$

(A) $\frac{1}{2}$

(B) 1

(C) $\ln 2$

(D) 2

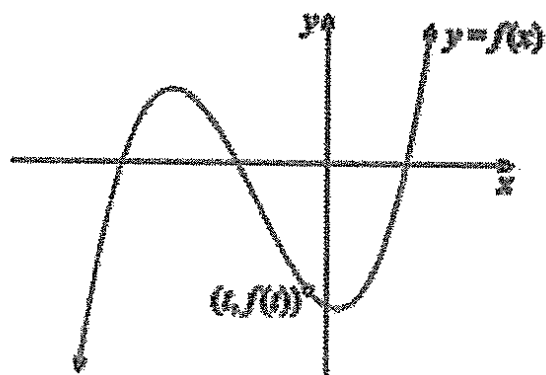
2. What is the derivative of $\log_2 x$?

(A) $\frac{1}{x}$

(B) $\frac{1}{2x}$

(C) $\ln 2x$

(D) $\frac{1}{x \ln 2}$

3. The diagram below shows the graph of $y = f(x)$. Which of the following statements is true?

(A) $f'(t) > 0$ and $f''(t) < 0$

(B) $f'(t) > 0$ and $f''(t) > 0$

(C) $f'(t) < 0$ and $f''(t) < 0$

(D) $f'(t) < 0$ and $f''(t) > 0$

4. Consider the curve given by $y = \frac{1}{2}x^4 - x^3$. What are the points of inflexion?

(A) only $(0, 0)$

(B) $(0, 0)$ and $(-1, \frac{5}{8})$

☒ (C) $(0, 0)$ and $(1, -\frac{1}{2})$

(D) $(0, 0)$ and $(\frac{3}{2}, -\frac{27}{32})$

5. The line $y = mx + b$ is a tangent to the curve $y = x^3 - 3x + 2$ at the point $(-2, 0)$. What is value of m and b ?

(A) $m = 9$ and $b = -18$

☒ (B) $m = 9$ and $b = 18$

(C) $m = 12$ and $b = -18$

(D) $m = 12$ and $b = 18$

Section II Answer Booklet

Student name

Student Number

Teacher Name

40 marks

Attempt Questions 6 - 14

Allow about 1 hour and 10 minutes for this section

Instructions

- Answer the questions in the spaces provided. Sufficient spaces are provided for typical responses.
- Your responses should include relevant mathematical reasoning and/or calculations.
- Extra writing space is provided at the back of the booklet. If you use this space, clearly indicate which question you are answering.

Question 6 (2 Marks)

Find the gradient of the tangent to the curve $y = \ln(\tan x)$ at the point on the curve where 2

$$x = \frac{\pi}{4}$$

Question 7 (9 Marks)

Consider the curve $y = f(x)$ where $f(x) = x^3 - 3x^2 - 9x + 27$

a) Show that $f(x) = (x+3)(x-3)^2$ 2

Question 7 continues on next page

b) Find the stationary points and determine their nature

3

c) Find the point of inflection

2

Question 7 continues on next page

d) Sketch the curve $y = f(x)$, showing all important features

2

Question 8 (3 Marks)

Consider the function $f(x) = x \ln x - x$

a) Find the values of x for which $f(x)$ is decreasing.

2

- b) Show that the curve $y = f(x)$ is concave up for all values of x in the domain of $f(x)$ 1
-

Question 9 (7 Marks)

Differentiate the following

a) $y = \tan(5x - 3)$ 1

b) $y = e^{2x} \sin x$ 2

c) $y = \frac{\tan x}{x}$ 2

Question 9 continues on next page

d) $y = \log_e \sqrt{x^2 - 1}$

2

Question 10 (3 Marks)

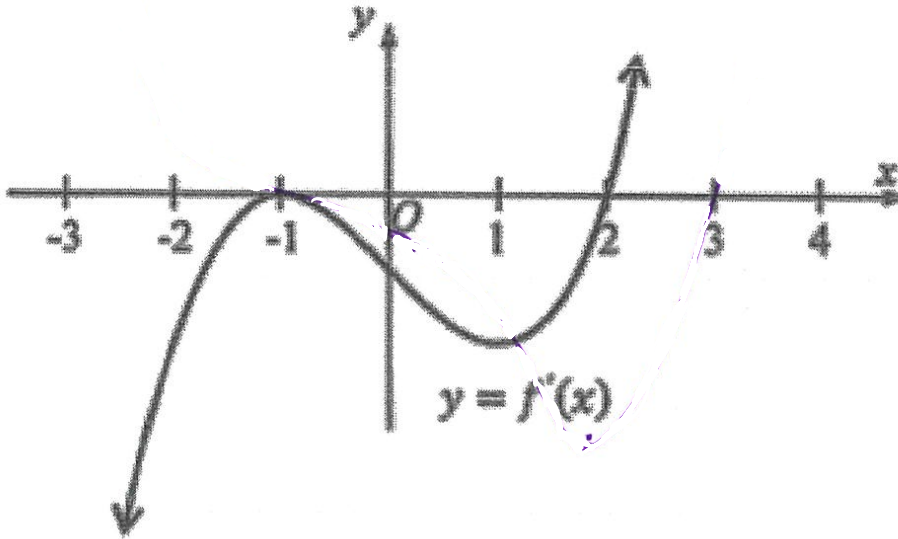
Find the value of m if $y = 2 \sin 3x + 4 \cos 2x$ and $\frac{d^2 y}{dx^2} = m \sin 3x - 4y$

3

Questions 6 - 10 are worth 24 marks in total

Question 11 (6 Marks)

The graph below is the curve $y = f'(x)$, the derivative of a function $f(x)$



a) For what values of x does $y = f(x)$ have stationary point?

1

b) For what values of x does $y = f(x)$ have a point of inflection?

1

c) For what values of x does $y = f(x)$ have a horizontal point of inflection?

1

d) Sketch a possible graph of $y = f(x)$ on the same set of axes given above.

3

Question 12 (3 Marks)

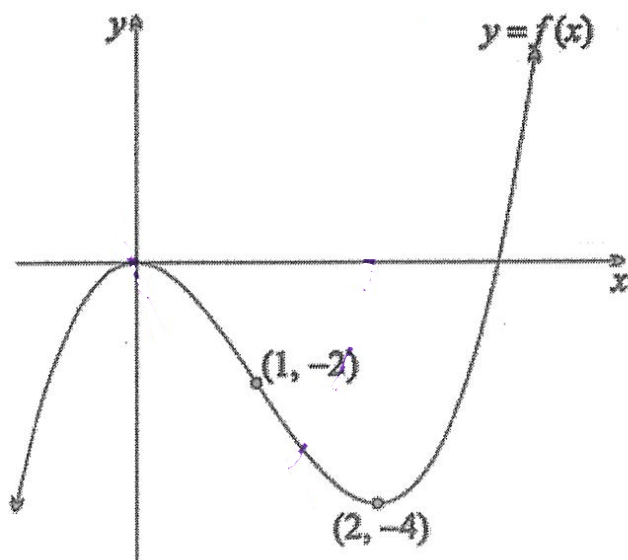
Find the equation of the tangent on $y = \log_e(2x^2 + 1)$ at the point $(2, \log_e 9)$.
Express your answer in general form.

3

Question 13 (2 Marks)

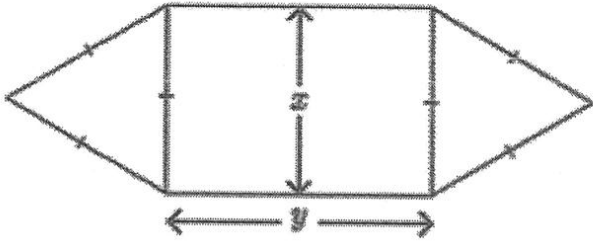
2

The function $y = f(x)$ is graphed in the diagram below. The points $(2, -4)$ and the origin are stationary points and the point $(1, -2)$ is a point of inflection. The gradient of the tangent at the point of inflection is -3 . Sketch the gradient function $y = f'(x)$ on the same set of axes. Mark on your sketch the co-ordinates of all intercepts with the axes and any stationary points.



Question 14 (5 Marks)

The figure shown below consists of a rectangle, x cm by y cm with two equilateral triangles on each side of length x cm. Given that the perimeter of the figure is 26 cm.



- a) Show that $A = 13x - \frac{(4 - \sqrt{3})x^2}{2}$ where A is the area of the figure

2

b) Hence show that the maximum area of the figure is $\frac{13(4+\sqrt{3})}{2}$

3

END OF EXAMINATION



Student No.

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(C) $\ln 2$

(D) 2

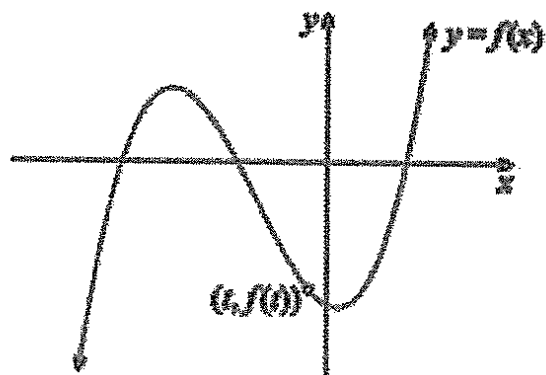
2. What is the derivative of $\log_2 x$?

(A) $\frac{1}{x}$

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3. The diagram below shows the graph of $y = f(x)$. Which of the following statements is true?

(A) $f'(t) > 0$ and $f''(t) < 0$

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(C) $f'(t) < 0$ and $f''(t) < 0$

(D) $f'(t) < 0$ and $f''(t) > 0$

4. Consider the curve given by $y = \frac{1}{2}x^4 - x^3$. What are the points of inflexion?

(A) only $(0, 0)$

(B) $(0, 0)$ and $(-1, \frac{5}{8})$

☒ (C) $(0, 0)$ and $(1, -\frac{1}{2})$

(D) $(0, 0)$ and $(\frac{3}{2}, -\frac{27}{32})$

5. The line $y = mx + b$ is a tangent to the curve $y = x^3 - 3x + 2$ at the point $(-2, 0)$. What is value of m and b ?

(A) $m = 9$ and $b = -18$

☒ (B) $m = 9$ and $b = 18$

(C) $m = 12$ and $b = -18$

(D) $m = 12$ and $b = 18$

Section II Answer Booklet

Student name

Student Number

Teacher Name

40 marks

Attempt Questions 6 - 14

Allow about 1 hour and 10 minutes for this section

Instructions

- Answer the questions in the spaces provided. Sufficient spaces are provided for typical responses.
- Your responses should include relevant mathematical reasoning and/or calculations.
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Question 6 (2 Marks)

Find the gradient of the tangent to the curve $y = \ln(\tan x)$ at the point on the curve where 2

$$x = \frac{\pi}{4}$$

$$y' = \frac{1}{\tan x} \times \sec^2 x$$

$$\text{At } x = \frac{\pi}{4}$$

$$y' = \frac{1}{\tan \frac{\pi}{4}} \times \sec^2 \frac{\pi}{4}$$

$$= (\sqrt{2})^2 = 2$$

Question 7 (9 Marks)

Consider the curve $y = f(x)$ where $f(x) = x^3 - 3x^2 - 9x + 27$ a) Show that $f(x) = (x+3)(x-3)^2$ 2

$$f(x) = x^3 - 3x^2 - 9x + 27$$

$$= x^2(x-3) - 9(x-3)$$

$$= (x^2 - 9)(x-3)$$

$$= (x-3)(x+3)(x-3)$$

$$= (x+3)(x-3)^2$$

Question 7 continues on next page

b) Find the stationary points and determine their nature

3

$$\begin{aligned}
 f'(x) &= 3x^2 - 6x + 9 \\
 &= 3(x^2 - 2x + 3) \\
 &= 3(x-3)(x+1)
 \end{aligned}$$

for stationary point $f'(x) = 0$

$$x = 3, -1$$

$$f''(x) = 6x - 6$$

$$\text{At } x = 3$$

$$f''(x) = 12$$

Minimum at $x = 3$

$$\text{At } x = -1$$

$$f''(x) = -12$$

$\therefore$ Maximum at $x = -1$

$$f(3) = 0, \quad f(-1) = 32$$

$\therefore (3, 0)$ is the min turning point.

$(-1, 32)$ is the max turning point.

c) Find the point of inflection

2

$$f''(x) = 6x - 6$$

$$6x - 6 = 0$$

$$x = 1$$

x	0	1	2
$f''(x)$	-6	0	6

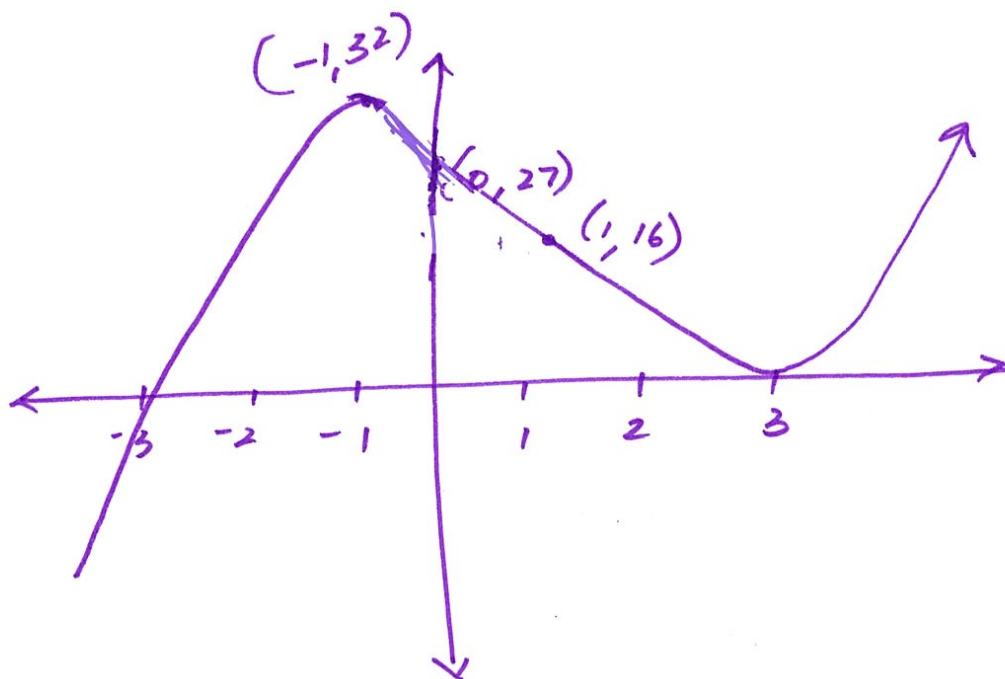
$f''(1) = 0$ $\times$ concavity changes

$\therefore (1, 16)$ is a point of inflexion

Question 7 continues on next page.

d) Sketch the curve $y = f(x)$, showing all important features

2



Question 8 (3 Marks)

Consider the function $f(x) = x \ln x - x$

a) Find the values of x for which $f(x)$ is decreasing.

2

$$\begin{aligned} f'(x) &= \left(\ln x + x \cdot \frac{1}{x} \right) - 1 \\ &= \ln x + 1 - 1 \\ &= \ln x \end{aligned}$$

$$f''(x) = \frac{1}{x}$$

$$f'(x) < 0$$

$$\ln(x) < 0$$

$$0 < x < 1$$

- b) Show that the curve $y = f(x)$ is concave up for all values of x in the domain of $f(x)$ 1

Domain is $x > 0$

$$f''(x) > 0 \text{ for } x > 0$$

curve is concave up through out the domain

Question 9 (7 Marks)

Differentiate the following

a) $y = \tan(5x - 3)$

1

$$y' = 5 \sec^2(5x - 3)$$

b) $y = e^{2x} \sin x$

2

$$\begin{aligned} y' &= e^{2x} \cos x + \sin x \times e^{2x} \times 2 \\ &= e^{2x} (\cos x + 2 \sin x) \end{aligned}$$

c) $y = \frac{\tan x}{x}$

2

$$\begin{aligned} y' &= \frac{x(\sec^2 x) - 1 \cdot \tan x}{x^2} \\ &= \frac{x \sec^2 x - \tan x}{x^2} \end{aligned}$$

Question 9 continues on next page

d) $y = \log_e \sqrt{x^2 - 1}$

2

$$y' = \frac{1}{\sqrt{x^2 - 1}} \times \frac{1}{\cancel{\sqrt{x^2 - 1}}} \times \cancel{2x}$$

$$= \frac{x}{x^2 - 1}$$

Question 10 (3 Marks)

3

Find the value of m if $y = 2 \sin 3x + 4 \cos 2x$ and $\frac{d^2 y}{dx^2} = m \sin 3x - 4y$

$$y' = 6 \cos 3x - 8 \sin 2x$$

$$y'' = -18 \sin 3x - 16 \cos 2x$$

$$-18 \sin 3x - 16 \cos 2x = m \sin 3x - 4y$$

$$-18 \sin 3x - 16 \cos 2x = m \sin 3x - 4(2 \sin 3x + 4 \cos 2x)$$

$$-18 \sin 3x - 16 \cos 2x = m \sin 3x - 8 \sin 3x - 16 \cos 2x$$

$$-18 \sin 3x + 8 \sin 3x = m \sin 3x$$

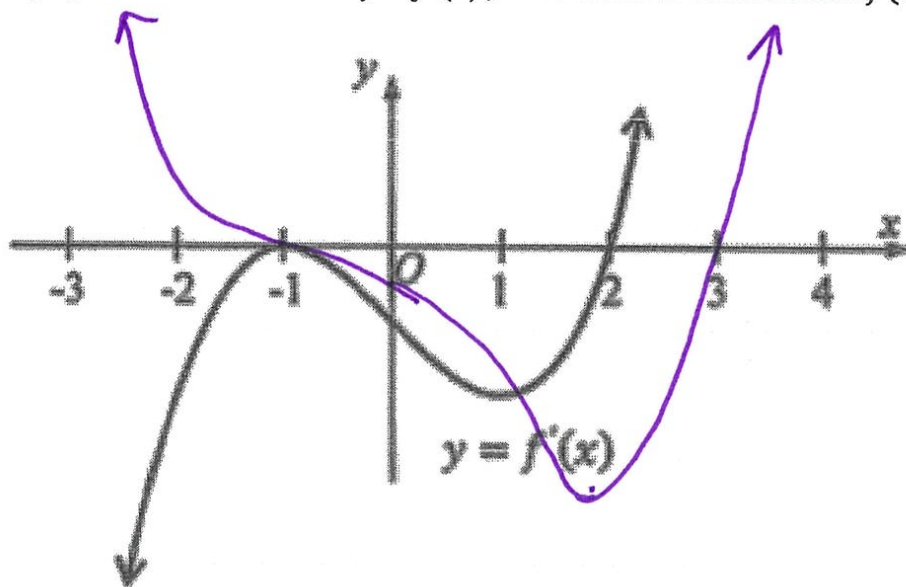
$$-10 \sin 3x = m \sin 3x$$

$$m = -10$$

Questions 6 - 10 are worth 24 marks in total

Question 11 (6 Marks)

The graph below is the curve $y = f'(x)$, the derivative of a function $f(x)$



- a) For what values of x does $y = f(x)$ have stationary point?

1

$$x = -1, 2$$

- b) For what values of x does $y = f(x)$ have a point of inflection?

1

$$x = -1 \text{ or } 1$$

- c) For what values of x does $y = f(x)$ have a horizontal point of inflection?

1

$$x = -1$$

- d) Sketch a possible graph of $y = f(x)$ on the same set of axes given above.

3

Question 12 (3 Marks)

Find the equation of the tangent on $y = \log_e(2x^2 + 1)$ at the point $(2, \log_e 9)$.
Express your answer in general form.

3

$$y = \ln(2x^2 + 1)$$

$$y' = \frac{1}{2x^2 + 1} \times 4x$$

$$\text{At } x=2, \quad m = \frac{4(2)}{2(2)^2 + 1} = \frac{8}{9}$$

$$y - \ln 9 = \frac{8}{9}(x - 2)$$

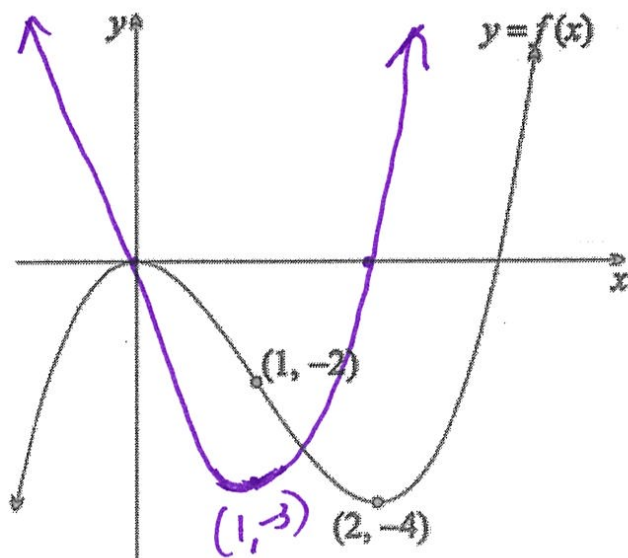
$$9y - 9\ln 9 = 8x - 16$$

$$8x - 9y - 16 + 9\ln 9 = 0$$

Question 13 (2 Marks)

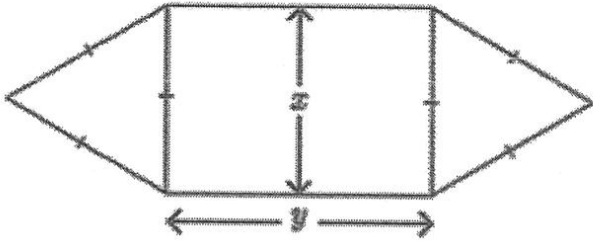
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The function $y = f(x)$ is graphed in the diagram below. The points $(2, -4)$ and the origin are stationary points and the point $(1, -2)$ is a point of inflection. The gradient of the tangent at the point of inflection is -3 . Sketch the gradient function $y = f'(x)$ on the same set of axes. Mark on your sketch the co-ordinates of all intercepts with the axes and any stationary points.



Question 14 (5 Marks)

The figure shown below consists of a rectangle, x cm by y cm with two equilateral triangles on each side of length x cm. Given that the perimeter of the figure is 26 cm.



- a) Show that $A = 13x - \frac{(4 - \sqrt{3})x^2}{2}$ where A is the area of the figure

2

$$2y + 4x = 26$$

$$y = \frac{26 - 4x}{2}$$

$$y = 13 - 2x$$

$$A_{\text{triangle}} = \frac{1}{2} x^2 \times \sin 60$$

$$= \frac{1}{2} x^2 \times \frac{\sqrt{3}}{2} = \frac{\sqrt{3}}{4} x^2$$

$$A_{\text{figure}} = \frac{\sqrt{3}}{4} x^2 \times 2 + xy$$

$$= \frac{\sqrt{3}}{2} x^2 + x(13 - 2x)$$

$$= \frac{\sqrt{3}}{2} x^2 + 13x - 2x^2$$

$$= 13x - 2x^2 + \frac{\sqrt{3}}{2} x^2$$

$$= 13x - \frac{4x^2}{2} + \frac{\sqrt{3}x^2}{2}$$

$$= 13x - \frac{x^2(4 - \sqrt{3})}{2}$$

b) Hence show that the maximum area of the figure is $\frac{13(4+\sqrt{3})}{2}$

3

$$\frac{dA}{dx} = 13 - \left(\frac{4-\sqrt{3}}{2} \right) \times 2x$$

$$= 13 - (4-\sqrt{3})x$$

For max value

$$\frac{dA}{dx} = 0$$

$$13 - (4-\sqrt{3})x = 0$$

$$(4-\sqrt{3})x = 13$$

$$x = \frac{13}{4-\sqrt{3}}$$

$$\frac{d^2A}{dx^2} = -(4-\sqrt{3}) < 0$$

$$\therefore \text{Maximum at } x = \frac{13}{4-\sqrt{3}}$$

$$\text{Maximum area} = 13 \left(\frac{13}{4-\sqrt{3}} \right) - \left(\frac{4-\sqrt{3}}{2} \right) \left(\frac{13}{4-\sqrt{3}} \right)^2$$

$$= \frac{13^2}{4-\sqrt{3}} - \frac{13^2}{2(4-\sqrt{3})}$$

$$= \frac{2 \times 13^2 - 13^2(4+\sqrt{3})}{2(4-\sqrt{3})(4+\sqrt{3})}$$

$$= \frac{13^2(4+\sqrt{3})}{2 \times 13} = \frac{13(4+\sqrt{3})}{2}$$

END OF EXAMINATION